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1  import java.awt.*;
2  import java.awt.event.*;
3  import java.net.MalformedURLException;
4  import java.nio.charset.MalformedInputException;
5  import java.rmi.*;
6  import java.rmi.server.RemoteObject;
7
8  import javax.swing.*;
9
10 public class GUI extends JFrame {
11     TextField name, email;
12     Choice pads;
13     Label message;
14
15     public GUI() {
16         setSize(300, 200);
17         setLayout(new GridLayout(6, 2));
18         add(new Label(" Name : "));
19         name = new TextField(30);
20         add(name);
21         add(new Label(" Email : "));
22         email = new TextField(30);
23         add(email);
24         add(new Label(" Pad : "));
25         pads = new Choice();
26         pads.addItem("Pad1");
27         pads.addItem("Pad2");
28         add(pads);
29         add(new Label(""));
30         add(new Label(""));
31         Button Abutton = new Button("add");
32         Abutton.addActionListener(new AButtonAction());
33         add(Abutton);
34         Button Cbutton = new Button("consult");
35         Cbutton.addActionListener(new CButtonAction());
36         add(Cbutton);
37         message = new Label();
38         add(message);
39     }
40
41     class CButtonAction implements ActionListener {
42         public void actionPerformed(ActionEvent ae) {
43             String n, c;
44             n = name.getText();
45             c = pads.getSelectedItem();
46             message.setText("consult(" + n + "," + c + ")");
47
48             try {
49                 Pad pad = (Pad)Naming.lookup("//localhost:8080/" + c);
50                 RRecord record = pad.consult(n, true);
51                 message.setText("Nom : "
52                     + record.getName()
53                     + " | Email : "

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54         + record.getEmail());
55     } catch (NotBoundException | MalformedURLException exc) {
56         exc.printStackTrace();
57     } catch (RemoteException exc) {
58         message.setText("Nom non trouvé !");
59     }
60 }
61 }
62
63 class AButtonAction implements ActionListener {
64     public void actionPerformed(ActionEvent ae) {
65         String n, e, c;
66         n = name.getText();
67         e = email.getText();
68         c = pads.getSelectedItem();
69         message.setText("add(" + n + ", " + e + ", " + c + ")");
70
71         try {
72             Pad pad = (Pad)Naming.lookup("//localhost:8080/"+c);
73             pad.add(new SRecordImpl(n, e));
74         } catch (RemoteException | NotBoundException | MalformedURLException exc) {
75             exc.printStackTrace();
76         }
77     }
78 }
79
80 public static void main(String args[]) {
81     GUI s = new GUI();
82     s.setSize(400, 200);
83     s.setVisible(true);
84 }
85 }
86
87 import java.rmi.*;
88
89 public interface Pad extends Remote {
90     public void add(SRecord sr) throws RemoteException;
91
92     public RRecord consult(String n, boolean forward) throws RemoteException;
93 }
94
95 import java.rmi.RemoteException;
96 import java.rmi.server.UnicastRemoteObject;
97 import java.util.HashMap;
98
99 public class PadImpl extends UnicastRemoteObject implements Pad {
100
101     private HashMap<String, String> pad;
102
103     public PadImpl() throws RemoteException {
104         this.pad = new HashMap<String, String>();
105     }
106
107     public void add(SRecord sr) throws RemoteException {

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108         this.pad.put(sr.getName(), sr.getEmail());
109     }
110
111     public RRecord consult(String n, boolean forward) throws RemoteException {
112         if (this.pad.containsKey(n)) {
113             return new RRecordImpl(n, this.pad.get(n));
114         } else {
115             throw new RemoteException("L'email n'existe pas.");
116         }
117     }
118 }
119
120 import java.rmi.*;
121
122 public interface RRecord extends Remote {
123     public String getName() throws RemoteException;
124     public String getEmail() throws RemoteException;
125 }
126
127 import java.io.Serializable;
128 import java.rmi.RemoteException;
129 import java.rmi.server.UnicastRemoteObject;
130
131 public class RRecordImpl extends UnicastRemoteObject implements RRecord {
132
133     private String name;
134     private String email;
135
136     public RRecordImpl(String name, String email) throws RemoteException {
137         this.name = name;
138         this.email = email;
139     }
140
141     public String getName() throws RemoteException {
142         return name;
143     }
144
145     public String getEmail() throws RemoteException {
146         return email;
147     }
148 }
149
150 import java.rmi.Naming;
151 import java.rmi.registry.LocateRegistry;
152
153 public final class Server {
154
155     public static final String serverURI = "///localhost:8080";
156
157     public static void main(String args[]) {
158         try {
159             LocateRegistry.createRegistry(8080);
160             PadImpl pad1 = new PadImpl();
161             PadImpl pad2 = new PadImpl();

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162         Naming.bind(serverURI+"/Pad1", pad1);
163         Naming.bind(serverURI+"/Pad2", pad2);
164     } catch (Exception e) {
165         e.printStackTrace();
166     }
167 }
168 }
169
170 import java.io.*;
171
172 public interface SRecord extends Serializable {
173     public String getName();
174     public String getEmail();
175 }
176
177 public class SRecordImpl implements SRecord {
178     private String name;
179     private String email;
180
181     public SRecordImpl(String name, String email) {
182         this.name = name;
183         this.email = email;
184     }
185
186     public String getName() {
187         return name;
188     }
189
190     public String getEmail() {
191         return email;
192     }
193 }
```